



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

FULL SYLLABUS

16-03-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [SCORE-I]

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	A	A	A	A	A	B	C	C
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	A	A	C	C	C	A	B	D	B

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	D	D	D	D	D	C	B	B
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	C	B	C	A	A	C	A	A	D

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	A	C	C	C	C	C	B	B
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	A	A	A	A	C	B	B	C	C

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. **Ans (B)**

$$t = \frac{d}{u - 10} = \frac{d}{10 - 5}$$

2. **Ans (C)**

$$N = mg \cos\phi + F \sin\theta$$

5. **Ans (A)**For W_1

$$F_x = 2xy - 2y^2 = 2x - 2$$

$$2 \left(\frac{x^2}{2} - x \right)_0^4 = 8J$$

For W_2

$$F_y = -2xy + 2x^2 = -8y + 32$$

$$W_2 = \int_1^3 (32 - 8y) dy$$

$$= 32 \times 2 - 4 \times (9 - 1) = 32J$$

For W_3

$$F_x = 6x^1 - 18$$

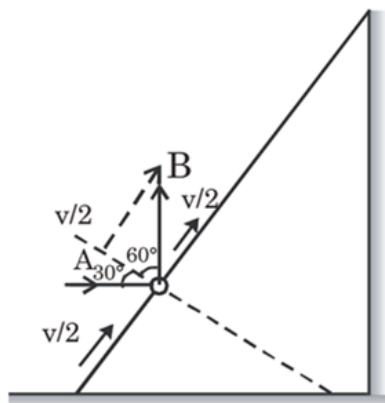
$$W_3 = \int_4^0 6x dx - 18 \int_4^0 dx$$

$$= -3 \times 16 + 18 \times 4 = 24J$$

For W_4

$$F_y = 0 \Rightarrow W_4 = 0$$

6. Ans (A)



Let 'v' be the initial velocity.

Tangential velocity remains same during collision and equal to

$$v \cos 60^\circ = v/2$$

Let v' be the normal component of velocity after impact.

$$\text{In } \triangle OAB : \tan 60^\circ = \frac{v/2}{v'} \Rightarrow v' = \frac{v}{2\sqrt{3}}$$

$$\text{Then : } e = \frac{v'}{v \cos 30^\circ} = \frac{(v/2\sqrt{3})}{(\sqrt{3}v/2)} = \frac{1}{3}$$

7. Ans (A)

This problem can be solved by analogy with linear SHM.

$$V^2 = \omega^2 (A^2 - x^2)$$

$$\left(\frac{3}{2}V\right)^2 = \omega^2 (A'^2 - x^2)$$

$$\text{Here, } x = A/2$$

$$\Rightarrow A' = \frac{A}{4}\sqrt{31}$$

$$\text{Hence } \theta' = \frac{\theta_0}{4}\sqrt{31}$$

8. Ans (B)

$$\vec{\tau} = \vec{P} \times \vec{E}$$

10. Ans (C)

$$\text{Initially } \frac{4}{Y} = \frac{40}{60}$$

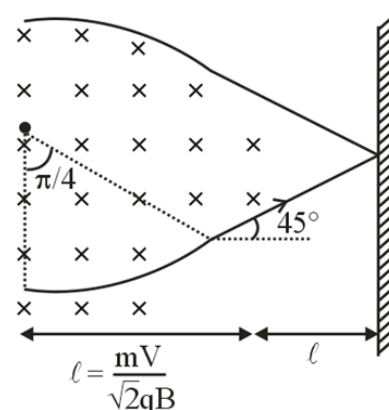
$$\text{So } Y = 60$$

$$\frac{Y' \times 78}{Y' + 78} = 6 \text{ again}$$

$$72Y' = 78 \times 6$$

$$Y' = \frac{78}{72} \times 6 = 6(1 + \alpha\Delta T)$$

11. Ans (A)



$$T = 2 \left(\frac{\pi}{4\omega} + \frac{\ell}{v \cos 45^\circ} \right)$$

$$\text{where } \omega = \frac{qB}{m}$$

12. Ans (A)

$$a = \frac{qE_0}{m} \quad T = \frac{2\pi m}{qB_0}$$

$$\Rightarrow S = \frac{1}{2}aT^2$$

$$\Rightarrow S = \frac{2\pi^2 m E_0}{qB_0^2}$$

13. Ans (A)

$$I = \frac{V}{\sqrt{R^2 + X_C^2}} \dots (A)$$

$$\text{If } \omega \rightarrow \frac{\omega}{3}, X_C \rightarrow 3X_C$$

$$\frac{I}{2} = \frac{V}{\sqrt{R^2 + 9X_C^2}} \dots (B)$$

From eq. (A) & (B)

$$\frac{X_C}{R} = \sqrt{\frac{3}{5}}$$

14. Ans (C)

$$\Delta\phi = 2n\pi \Rightarrow \frac{\pi}{2} + \frac{2\pi}{\lambda}d \sin \theta = 2n\pi ;$$

$$\frac{2\pi}{\lambda}d \sin \theta = \left(2n - \frac{1}{2}\right)\pi$$

$$\sin \theta = \left(2n - \frac{1}{2}\right) \frac{\lambda}{2d} = \frac{1}{2} \times \frac{\lambda}{2 \times 3\lambda} = \frac{1}{12}$$

$$\Rightarrow \frac{y}{\sqrt{(100\lambda)^2}} = \frac{1}{2}$$

$$144y^2 = (100\lambda)^2; y \approx \frac{100\lambda}{12} = \frac{25\lambda}{3}$$

16. Ans (C)

$$\lambda_{(\alpha+\beta)} = \lambda_{\alpha} + \lambda_{\beta}$$

$$\Rightarrow \frac{1}{T_{\frac{1}{2}(\alpha+\beta)}} = \frac{1}{T_{\frac{1}{2}(\alpha)}} + \frac{1}{T_{\frac{1}{2}(\beta)}}$$

$$\Rightarrow \frac{1}{T_{\frac{1}{2}(\alpha+\beta)}} = \frac{1}{30} + \frac{1}{60} = \frac{1}{20}$$

$$\therefore T_{\frac{1}{2}(\alpha+\beta)} = 20 \text{ years}$$

$\therefore$ One-fourth of sample will remain after 2 half life = 40 years.

18. Ans (B)

$$50 \text{ divisions} = 2.45 \text{ cm}$$

$$\Rightarrow 1 \text{ division} = \frac{2.45}{50} = 0.049 \text{ cm}$$

$$\Rightarrow \text{least count} = 0.05 - 0.049 = 0.001 \text{ cm}$$

$$\text{So vernier reading} = 0.001 \times 24 = 0.024 \text{ cm}$$

$$\Rightarrow \text{diameter of cylinder} = 5.10 + 0.024 = 5.124 \text{ cm}$$

20. Ans (B)

$$f_0 = 80 \text{ cm}$$

$$f_e = 5 \text{ cm}$$

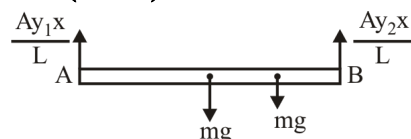
$$\ell = f_0 + f_e = 85 \text{ cm}$$

$$m = \frac{f_0}{f_e} = 16$$

PART-1 : PHYSICS

SECTION-II

1. Ans (4000)



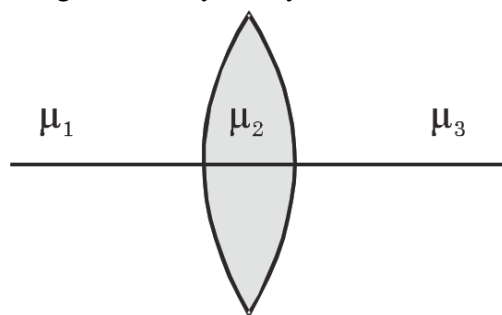
$$\frac{Ax}{L} (Y_1 + Y_2) = (M + m)g$$

$$\left(\frac{Ay_1x}{L} \right) \frac{\ell}{2} + mg \times \frac{\ell}{4} = \left(\frac{Ay_2x}{L} \right) \frac{\ell}{2}$$

$$\frac{Y_1}{Y_2} = \frac{2M + m}{2M + 3m}$$

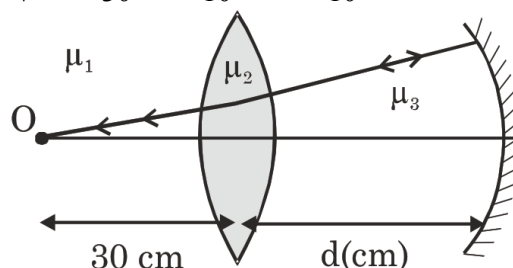
2. Ans (7)

Image formed by this system is



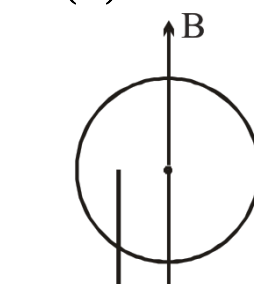
$$\frac{\mu_3}{v} - \frac{\mu_1}{u} = \frac{\mu_3 - \mu_1}{R_1} + \frac{\mu_1 - \mu_2}{R_2}$$

$$\frac{4}{v} - \frac{1}{-30} = \frac{2-1}{10} + \frac{4-2}{-10}; v = -30$$



$$d + 30 = 100, d = 70 \text{ cm}$$

3. Ans (4)

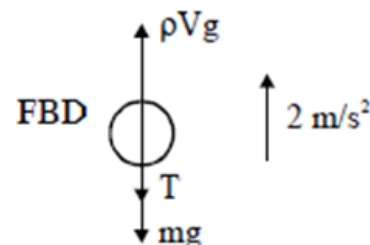


$$6\pi\eta r v$$

$$1000 \times \frac{4}{3} \pi r^3 g - 500 \times \frac{4}{3} \pi r^3 g - 6\pi\eta r v$$

$$= 500 \times \frac{4}{3} \pi r^3 a$$

4. Ans (16)



$$\rho Vg - mg - T = ma.$$

$$\text{But } \rho Vg = 2mg;$$

$$\text{so } T = m(g - a) = 2(10 - 2) = 16\text{N}$$

5. **Ans (4)**

$$\text{Area of DABE} = \frac{1}{2} (2.5 \times 10^5) (0.05 \times 10^{-3})$$

$$= \frac{125}{2} = \frac{25}{4}$$

since, DAEB and DBCD are similar, therefore

$$\frac{1.5}{2.5} = \frac{CD}{0.05}$$

$$CD = \frac{3}{5} \times 0.05 = 0.03$$

$$\text{Area of DBCD} = \frac{1}{2} \times (1.5 \times 10^5) (0.03 \times 10^{-3})$$

$$= \frac{4.5}{2} = \frac{9}{4}$$

$$W_N = \frac{25}{4} - \frac{9}{4} = 4J$$

6. **Ans (8)**

Conserving angular momentum :

$$m(v_1 \cos 60^\circ) 4R = mv_2 2R \Rightarrow \frac{v_2}{v_1} = 2$$

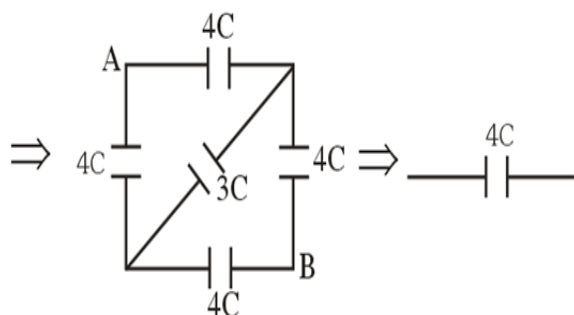
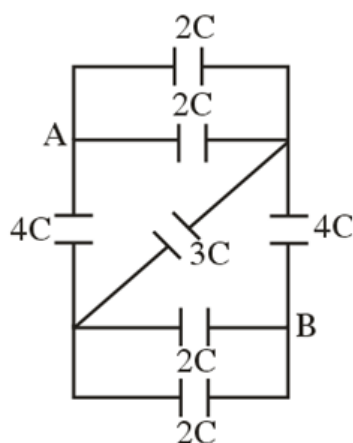
Conserving energy of the system

$$-\frac{GMm}{4R} + \frac{1}{2}mv_1^2 = -\frac{GMm}{R} + \frac{1}{2}mv_2^2$$

$$\frac{1}{2}v_2^2 - \frac{1}{2}v_1^2 = \frac{3}{4} \frac{GM}{R} \Rightarrow v_1^2 = \frac{1}{2} \frac{GM}{R} ;$$

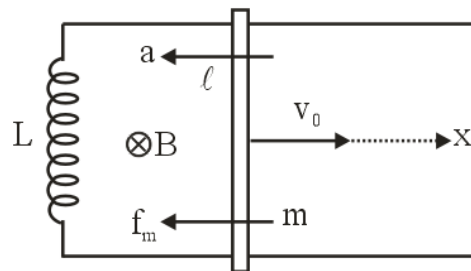
$$v_1 = \frac{1}{\sqrt{2}} \sqrt{64 \times 10^6} = \frac{8000}{\sqrt{2}} \text{ m/s} \Rightarrow X = 8$$

7. **Ans (4)**

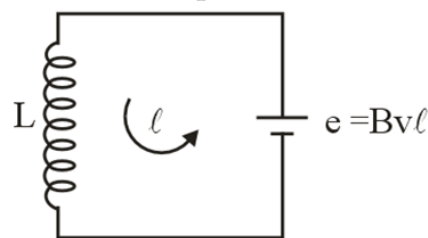


8. **Ans (5)**

According to Len's law, magnetic force, will oppose the motion of conducting rod.



electrical equivalent



$$\text{Here, } Bv_l = L \frac{di}{dt} \Rightarrow B(v dt)l = L(di) \Rightarrow B l (dx)$$

$$= L(di) \Rightarrow x = \frac{di}{Bl}$$

Force =

$$i l B = \frac{B^2 l^2}{L} x = m \omega^2 x \Rightarrow \omega = \frac{Bl}{\sqrt{mL}}$$

$$\text{So amplitude } \frac{v_0}{\omega} = \frac{v_0}{Bl} \sqrt{mL} = 5 \text{ cm}$$

9. **Ans (8)**

Speed of pulse in lower and upper parts are $2V$ & V respectively. The lower pulse will travel the

$$\text{complete length } \ell/2 \text{ in } t_1 = \frac{\ell/2}{2V} = \frac{\ell}{4V} \text{ sec. In this}$$

time upper pulse travels a distance $d = Vt_1 = \ell/4$.

Now the remaining distance that both pulses have to travel with equal speeds is $\frac{3\ell}{4}$. Hence both will travel distance.

$\therefore$ Required distance from upper end

$$= \frac{\ell}{4} + \frac{3\ell}{8} = \frac{5\ell}{8}$$

10. Ans (2)

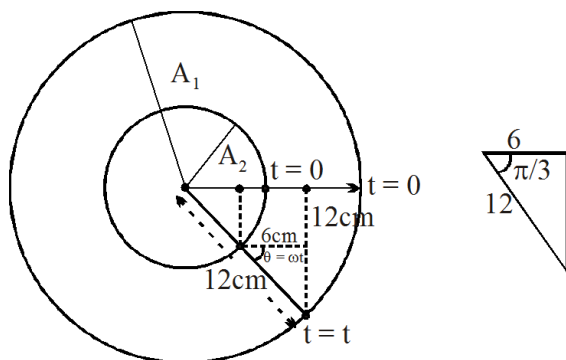
$$x_1 = A_1 \cos(\omega t)$$

$$x_2 = A_2 \cos(\omega t)$$

$$x_1 - x_2 = (A_1 - A_2) \cos(\omega t)$$

$$6 = 12 \cos(\omega t)$$

$$\omega t = \frac{\pi}{3} \Rightarrow t = \frac{\pi}{3\omega} = \frac{T}{6} = 2s$$



Alternative :

$$\theta = \frac{\pi}{3}; \omega t = \frac{\pi}{3}$$

$$t = \frac{T}{6} = 2 \text{ sec.}$$

PART-2 : CHEMISTRY

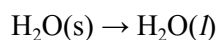
SECTION-I

8. Ans (C)

In electro-osmosis, the sol particles are prevented from migration whereas the dispersion medium migrates in the direction opposite to those of particles. Here, the medium is negatively charged.

9. Ans (B)

Reaction will be spontaneous at 400 K and 1 atm



$$\Delta S = 60 - 38 = 22 \text{ Jmol}^{-1}\text{K}^{-1}$$

$$\Delta G < 0$$

$$\Delta H - T\Delta S < 0$$

$$\Delta H < T\Delta S$$

$$\Delta H < (400) (22)$$

$$\Delta H < 8800$$

10. Ans (B)

Antifluorite structure = Anion + Cation

FCC lattice point Tetrahedral voids
 S^{2-} $\text{K}^{\oplus}$

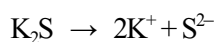
No. of molecules in an unit cell = 4

No. of unit cell = 30.11×10^{22} unit cell

Total number of molecules = $30.11 \times 4 \times 10^{22}$ molecules

$$\text{Moles of molecules} = \frac{30.11 \times 4 \times 10^{22}}{6.022 \times 10^{23}} = 2$$

Van't Hoff factor = $i = 3$ as it is strong electrolyte.



$$\Delta T_f = K_f m_i$$

$$\Delta T_f = 1.86 \times 0.2 \times 3 \quad 10 \text{ litre of water} = 10 \text{ kg of water}$$

$$\Delta T_f = 1.116 \quad \text{Density} = 1 \text{ g/mL} = 1 \text{ kg/litre}$$

$$273 - T_f = 1.116 \quad \text{Molality} = \frac{2}{10} = 0.2$$

$$T_f = 271.884$$

11. Ans (C)

Given : ΔH when a solution is diluted from 4m to 2m is -1.6 kJ/mole

4M means that 4 moles of solute is present in one litre of solution.

On dilution the no. of moles of solute remains constant

$\therefore$ In 5 litre solution, the no of moles present = $5 \times 4 = 20$ moles

$\therefore \Delta H$ when 5 litre of such solution is diluted will be

$$= -1.6 \times 20$$

$$= -32 \text{ kJ}$$

12. Ans (C)

$$\Delta G = \Delta H - T\Delta S$$

13. Ans (B)

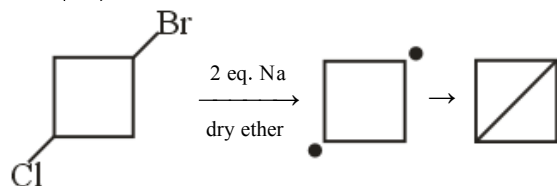
$$\frac{r_{\text{O}_2}}{r_{\text{gas}}} = \sqrt{\frac{3}{2}} = \sqrt{\frac{M_{\text{gas}}}{32}} \Rightarrow M_{\text{gas}} = 48$$

$$V_{m, \text{real}} = \frac{1000}{800} \times 48 \text{ l} = 60 \text{ l}$$

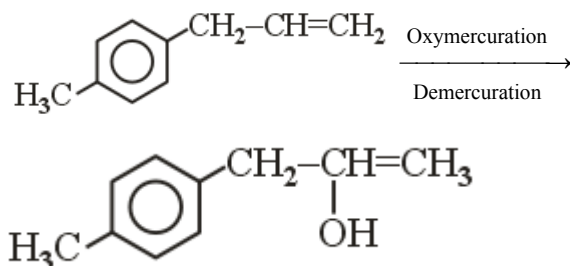
$$V_{m, \text{ideal}} = \frac{RT}{P} = \frac{0.08 \times 500}{1} = 40 \text{ l}$$

$$Z = \frac{60}{40} = \frac{3}{2}$$

14. Ans (C)



16. Ans (A)



17. Ans (C)

Rearrangement take place via cyclic transition formation so position 3 isotopically C-atom is present

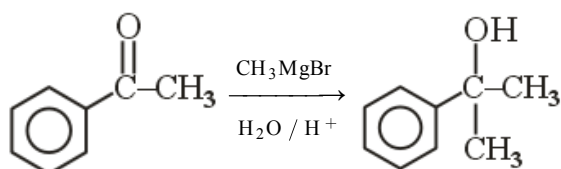
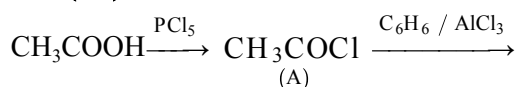
18. Ans (A)

Natural rubber is polymer of Cis-1,4-polyisoprene

19. Ans (A)

Sucrose formed by glycosidic link of C-1 α -glucose, C-2 of β -fructose

20. Ans (D)



PART-2 : CHEMISTRY

SECTION-II

4. Ans (2)

$$r = \frac{n^2 \cdot a_0}{Z}$$

from equation, $\frac{n^2 \times 0.529}{3} = 1.587 \Rightarrow n = 3$

Now, as $(n - l - 1) = 0 \Rightarrow l = 2$

5. Ans (3)

$$\ln \left[\frac{(t_{1/2})_1}{(t_{1/2})_2} \right] = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$\ln \left[\frac{3}{(t_{1/2})_2} \right] = \frac{840R}{R} \left(\frac{1}{300} - \frac{1}{400} \right)$$

$(t_{1/2})_2 = 1.5 \text{ min.}$

6. Ans (2700)

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}} = \left(\frac{32}{1} \right)^{\frac{2/3}{5/3}} = \left(\frac{32}{1} \right)^{2/5} = 4$$

$T_2 = 4 \times T_1 = 1200 \text{ K}$

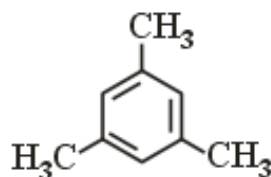
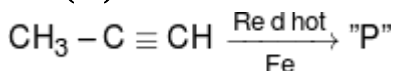
$\Delta U = nC_V \Delta T$

$= 1 \times \frac{3}{2} \times 2 \times 900 = 2700 \text{ cal.}$

7. Ans (4)

If n fused ring are connected then number of resonance is $(n + 1)$ n = no of ring

8. Ans (4)



9. Ans (3)

Aromatic compound 2, 4, 6

PART-3 : MATHEMATICS

SECTION-I

1. Ans (C)

The plane containing the given line is

$$(2x + 3y + 5z + 1) + \lambda(3x + 4y + 6z + 2) = 0$$

$$\Rightarrow (2 + 3\lambda)x + (3 + 4\lambda)y + (5 + 6\lambda)z + (1 + 2\lambda) = 0$$

This plane is parallel to y-axis

$$\Rightarrow 3 + 4\lambda = 0 \Rightarrow \lambda = -\frac{3}{4}$$

$\Rightarrow$ A point on y-axis is the origin.

$\Rightarrow$ perpendicular distance of the origin from the plane $x - 2z + 2 = 0$ is $\frac{2}{\sqrt{5}}$

2. Ans (B)

$$g(x) = x^2 + \frac{1}{x^2}$$

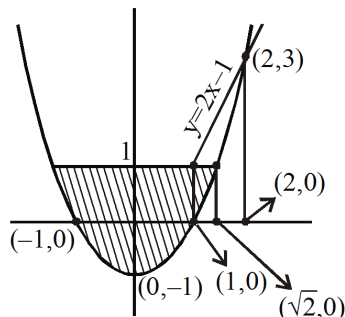
$$f(g(x)) = x^6 + \frac{1}{x^6} = \left(x^2 + \frac{1}{x^2}\right)^3 - 3\left(x^2 + \frac{1}{x^2}\right)$$

$$f(g(x)) = (g(x))^3 - 3g(x)$$

$$\Rightarrow f(x) = x^3 - 3x$$

$$f'(x) = 3x^2 - 3$$

3. Ans (A)



$$y = \begin{cases} x - 1 + x; & x > 1 \\ -x + 1 + x; & x \leq 1 \end{cases}$$

$$= \begin{cases} 2x - 1; & x > 1 \\ 1; & x \leq 1 \end{cases}$$

Required area

$$= \int_{x=-\sqrt{2}}^{x=\sqrt{2}} [1 - (x^2 - 1)] dx + \int_{y=1}^{y=3} \left[\sqrt{y+1} - \frac{y+1}{2} \right] dy = \frac{7+4\sqrt{2}}{3}$$

4. Ans (C)

General term $= {}^{1024}C_r (5^{1/2})^{1024-r} ((7)^{1/8})^r$ will be integer. if r is multiple of 8

$$\Rightarrow r = 0, 8, 16, 24, \dots, 1024$$

$$\text{sum} = 8 \times 128 = 1024$$

$$\Rightarrow \text{Number of integral terms} = 129.$$

5. Ans (C)

$$S_1 : (x - 1)^2 + (y - 1)^2 + \lambda(x + y - 2) = 0$$

$$S_2 : x^2 + y^2 + 4x + 5y - 4 = 0$$

$$\text{chord PQ} : S_1 - S_2 = 0$$

$$(-6x - 7y + 6) + \lambda(x - y - z) = 0$$

$$\Rightarrow x = 8, y = -6$$

7. Ans (C)

$$f(x) = |\cos x| + \cos^{-1}(\text{sgn} x) + |\ln x|$$

$$f(x) = |\cos x| + \cos^{-1}(1) + |\ln x| \quad \forall x \in (0, 2\pi)$$

$$= |\cos x| + |\ln x| \quad \forall x \in (0, 2\pi)$$

$$\therefore f(x) \text{ is not differentiable at } x = 1, \frac{\pi}{2}, \frac{3\pi}{2}$$

8. Ans (C)

$$y^2 = 2cx + 2c^{3/2} \dots\dots\dots(1)$$

$$2y \frac{dy}{dx} = 2c$$

$$y \frac{dy}{dx} = c \dots\dots\dots(2)$$

from (1) and (2)

$$y^2 = 2y \frac{dy}{dx} (x) + 2 \left(\frac{y dy}{dx} \right)^{3/2}$$

$$\left(y - 2x \frac{dy}{dx} \right) = 2 \left(\frac{dy}{dx} \right)^{3/2} y^{1/2}$$

on squaring

$$\text{degree} = 3 \quad \text{order} = 1$$

9. Ans (B)

Here $|A| = 4$ where $A = \begin{bmatrix} x & 1 & 1 \\ 2 & x & 3 \\ x & -1 & 2 \end{bmatrix}$, then

$$\begin{vmatrix} 2x+3 & -3 & 3-x \\ 3x-4 & x & 2-3x \\ -x^2-2 & 2x & x^2-2 \end{vmatrix} = |\text{adj} A| = |A|^2 = 16$$

10. Ans (B)

Let middle point of chord be (h, k)

$$T = S_1 \Rightarrow \frac{xh}{3} + \frac{yk}{4} = \frac{h^2}{3} + \frac{k^2}{4}$$

Chord passing through $(2, 0)$

$$\Rightarrow \frac{2h}{3} = \frac{h^2}{3} + \frac{k^2}{4}$$

$$\Rightarrow \frac{(h-1)^2}{1} + \frac{k^2}{4/3} = 1 \Rightarrow \frac{(x-1)^2}{1} + \frac{y^2}{4/3} = 1$$

11. Ans (C)

$$\lim_{x \rightarrow 0} \frac{f(x^2 + x + 1) - f(1)}{f(x^4 - x^2 + 2x + 4) - f(4)}$$

$$\left(\frac{0}{0} \text{ form} \right)$$

As. $f(x)$ is differentiable then apply

L' Hospital rule

$$= \lim_{x \rightarrow 0} \frac{(2x + 1)f'(x^2 + x + 1)}{(4x^3 - 2x + 2)f'(x^4 - x^2 + 2x + 4)}$$

$$= \frac{f'(1)}{2f'(4)} = 4$$

12. Ans (A)

$$\therefore VA = 0 \text{ (Null matrix)}$$

$$\Rightarrow (\log a)x + (\log b)y + (\log c)z = 0$$

$$(\log b)x + (\log c)y + (\log a)z = 0$$

$$(\log c)x + (\log a)y + (\log b)z = 0$$

which is homogenisis system of linear equations in x, y and z .

$$\therefore x, y, z \neq 0$$

$\Rightarrow$ system possesses non trivial solutions.

$$\Rightarrow D = \begin{vmatrix} \log a & \log b & \log c \\ \log b & \log c & \log a \\ \log c & \log a & \log b \end{vmatrix} = 0$$

$$\Rightarrow \log a + \log b + \log c = 0$$

$$\text{OR } \log a + \log b + \log c$$

$$\therefore a, b, c \text{ are distinct}$$

$$\Rightarrow \log a + \log b + \log c = 0$$

$$\Rightarrow abc = 1$$

$$\Rightarrow b^3 = 1 \quad (\because b^2 = ac)$$

$$\Rightarrow b = 1 \Rightarrow ac = 1 \Rightarrow c = \frac{1}{a}$$

$\Rightarrow$ System become

$$(\log a)x - (\log a)z = 0$$

$$-(\log a)y + (\log a)z = 0$$

$$-(\log a)x - (\log a)y = 0$$

$$\Rightarrow x - z = 0$$

$$y - z = 0$$

$$x - y = 0$$

$$\Rightarrow x = y = z$$

$$\Rightarrow \text{solutions lie on } \frac{x}{1} = \frac{y}{1} = \frac{z}{1}$$

$$\Rightarrow \alpha : \beta : \gamma = 1 : 1 : 1$$

$$\frac{\alpha}{\beta} + \frac{\beta}{\gamma} + \frac{\gamma}{\alpha} = 3$$

13. Ans (A)

$$|\text{adj}(\text{adj} A)| = |A|^{(n-1)^2}$$

$$|\text{adj}(\text{adj}(\text{adj}(A)))| = |A|^{(n-1)^3} = 3^8$$

14. Ans (A)

$$\ell n(\ell nx - \ell ny) = e^{x^2y} (1 - \ell nx) \quad \dots(1)$$

diff. w.r. to x .

$$\frac{1}{(\ell nx - \ell ny)} \cdot \left(\frac{1}{x} - \frac{1}{y} \frac{dy}{dx} \right) = e^{x^2y} \left(-\frac{1}{x} \right) + (1 - \ell nx) e^{x^2y} \left(x^2 \frac{dy}{dx} + y \cdot 2x \right) \quad \dots(2)$$

$$\text{Put } x = e \text{ in (1), (1)} \Rightarrow \ell n(1 - \ell ny) = 0 \Rightarrow y = 1$$

Now put $x = e, y = 1$ in (2) then (2)

$$\Rightarrow \left(\frac{1}{e} - y'(e) \right) = -\frac{e^{e^2}}{e}$$

$$\Rightarrow y'(e) = \frac{1 + e^{e^2}}{e}$$

15. Ans (A)

$$\underbrace{A, I, A, I, O}_5 \text{ \&P, P, L, C, T, N}_6$$

$$\text{ways} = \frac{6!}{2!} \cdot {}^7C_5 \cdot \frac{5!}{2!2!} = (45) 7!$$

16. Ans (C)

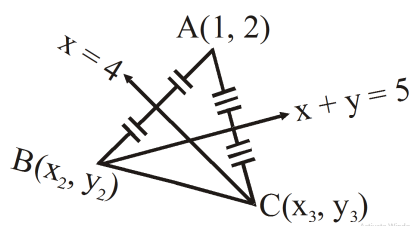
$$\text{Let } h = 2t_1^2 \text{ \& } 12 = 4t_2$$

$$\text{We know } t_2 = -t_1 - \frac{2}{t_1}$$

$$\Rightarrow t_1 = -1 \text{ or } -2$$

$$\Rightarrow h = 2 \text{ or } 8$$

17. Ans (B)



$$\frac{x_2 + 1}{2} = 4$$

$$\Rightarrow x_2 = 7$$

$$x_2 + y_2 = 5$$

$$\Rightarrow 7 + y_2 = 5$$

$$\Rightarrow y_2 = -2$$

$$\therefore (x_2, y_2) \equiv (7, -2)$$

$$x_3 = 4$$

$$\frac{x_3 + 1}{2} + \frac{y_3 + 2}{2} = 5$$

$$\Rightarrow 4 + y_3 = 10 - 3 = 7$$

$$\Rightarrow y_3 = 3$$

$$\therefore (x_3, y_3) \equiv (4, 3)$$

18. Ans (B)

$$S = \frac{1}{1^5} + \frac{1}{3^5} + \frac{1}{5^5} + \frac{1}{7^5} + \dots$$

$$= \left(\frac{1}{1^5} + \frac{1}{2^5} + \frac{1}{3^5} + \frac{1}{4^5} + \dots \right) - \left(\frac{1}{2^5} + \frac{1}{4^5} + \frac{1}{6^5} + \dots \right)$$

$$= \text{(k)} - \frac{1}{32} \left(\frac{1}{1^5} + \frac{1}{2^5} + \frac{1}{3^5} + \dots \right)$$

$$= k - \frac{k}{32} = \frac{31k}{32}$$

19. **Ans (C)**

Let $x_i - 5 = d_i$

$$\sigma_x^2 = \sigma_d^2 = \frac{\sum d_i^2}{n} - \left(\frac{\sum d_i}{n} \right)^2$$

$$= \frac{125}{10} - \left(\frac{5}{10}\right)^2 = \frac{25}{2} - \frac{1}{4} = \frac{49}{4}$$

20. Ans (C)

focus is foot of perpendicular drawn from vertex on latus rectum i.e. (4,3) and equation of the directrix is $x + y + c = 0$

$$\text{i.e. } \left| \frac{1+c}{\sqrt{2}} \right| = 3\sqrt{2} \Rightarrow c = 5 \text{ or } -7$$

$\therefore$ equation of the directrix is $x + y + 5 = 0$

$\therefore$ equation of the required parabola is

$$(x-4)^2 + (y-3)^2 = \left(\frac{x+y+5}{\sqrt{2}}\right)^2$$

$$\Rightarrow x^2 + y^2 - 2xy - 26x - 22y + 25 = 0$$

Compare with $x^2 + y^2 - 2xy + px + qy + r = 0$

$$\Rightarrow p = -26, q = -22, r = 25$$

$$\therefore p + q + r = -23$$

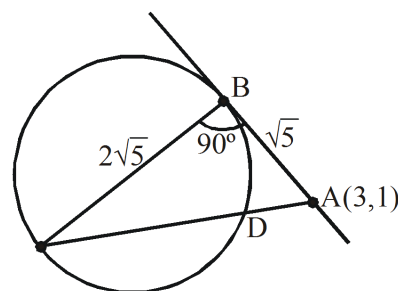
PART-3 : MATHEMATICS

SECTION-II

1. Ans (2)

$$\begin{vmatrix} a & x & 0 \\ b & y & 0 \\ c & z & 0 \end{vmatrix} \times \begin{vmatrix} x & a & b \\ y & b & c \\ z & c & a \end{vmatrix} + \begin{vmatrix} 1 & -1 & 1 \\ \cos \theta & \cos \theta & \sin \theta \\ -\sin \theta & -\sin \theta & \cos \theta \end{vmatrix} = 2$$

2. Ans (4)



$$\angle ABC = 90^\circ$$

$$\text{BC} = 2\sqrt{5}$$

$$AB = \sqrt{9 + 1 - 5} = \sqrt{5}$$

$$\Rightarrow AC = 5$$

Power of point $\Rightarrow AB^2 = AD.AC$

$$\Rightarrow AD = 1 \Rightarrow CD = 4$$

3. **Ans (11)**

$$\begin{aligned} y &= \sin^{-1}(\sin 6) - \tan^{-1}(\tan 8) + \cos^{-1}(\cos 6) \\ &= 6 - 2\pi - (8 - 3\pi) + 2\pi - 6 \\ &= 3\pi - 8 \\ a &= 3, b = -8 \end{aligned}$$

4. **Ans (8)**

$$\begin{aligned} I &= \int \frac{\sin^3 x dx}{1 + \sin^2 x} \\ \Rightarrow \text{let } \cos x &= t \Rightarrow \sin x dx = -dt \\ \Rightarrow I &= \int \frac{(1 - t^2) dt}{t^2 - 2} \\ &= \int \frac{(2 - t^2) - 1}{(t^2 - 2)} dt \\ &= -t - \frac{1}{2\sqrt{2}} \ln \left| \frac{t - \sqrt{2}}{t + \sqrt{2}} \right| \\ &= -\cos x - \frac{1}{2\sqrt{2}} \ln \left| \frac{\cos x - \sqrt{2}}{\cos x + \sqrt{2}} \right| \end{aligned}$$

5. **Ans (1)**

Equation of chord is

$$\frac{x - 5}{\cos \alpha} = \frac{y - 3}{\sin \alpha} = r$$

Point $(5 + r\cos\alpha, 3 + r\sin\alpha)$ of chord is one of its end point

$$2(5 + r\cos\alpha)^2 - 3(3 + r\sin\alpha)^2 = 6$$

$(5, 3)$ is midpoint of chord iff sum of roots = 0

$$20 \cos \alpha - 18 \sin \alpha = 0$$

$$\frac{\cos \alpha}{9} = \frac{\sin \alpha}{10} \Rightarrow \tan \alpha = \frac{10}{9} = m$$

$$\therefore p - q = 1$$

6. **Ans (192)**

$$\begin{aligned} T_n &= \frac{1}{n(n+2)(n+4)} \\ T_n &= \frac{1}{4} \left(\frac{1}{n(n+2)} - \frac{1}{(n+2)(n+4)} \right) \\ T_1 &= \frac{1}{4} \left(\frac{1}{1.3} - \frac{1}{3.5} \right) \\ T_2 &= \frac{1}{4} \left(\frac{1}{2.4} - \frac{1}{4.6} \right) \\ T_3 &= \frac{1}{4} \left(\frac{1}{3.5} - \frac{1}{5.7} \right) \quad \frac{22}{S} = 192 \\ S_\infty &= \frac{1}{4} \left(\frac{1}{3} + \frac{1}{8} \right) = \frac{11}{96} \end{aligned}$$

7. **Ans (5)**

$$\text{Let } \vec{c} = \lambda \vec{a} + \mu \vec{b}$$

take dot by $\vec{b}$

$$0 = \lambda (\vec{a} \cdot \vec{b}) + \mu (\vec{b} \cdot \vec{b}) = -\lambda + 5\mu$$

$$\Rightarrow \lambda - 5\mu = 0 \quad \dots(1)$$

$$\text{again } \vec{a} \cdot \vec{c} = 7 \Rightarrow \lambda \vec{a}^2 + \mu (\vec{a} \cdot \vec{b}) = 7$$

$$\Rightarrow 3\lambda - \mu = 7 \quad \dots(2)$$

$$\text{solving (1) and (2) } \lambda = \frac{5}{2}, \mu = \frac{1}{2}$$

$$\therefore \vec{c} = \frac{5}{2}(-\hat{i} + \hat{j} + \hat{k}) + \frac{1}{2}(2\hat{i} + \hat{k}) = -\frac{3\hat{i}}{2} + \frac{5\hat{j}}{2} + 3\hat{k}$$

$$\Rightarrow \frac{2}{7} |\vec{c}|^2 = \frac{1}{7} \times 35 = 5$$

8. **Ans (12)**

$$9\cos^{12}x + \cos^2 2x + 1 = 6\cos^6 x \cos 2x + 6\cos^6 x$$

$$-2\cos 2x = 0$$

$$(3\cos^6 x - 1 - \cos 2x)^2 = 0$$

$$3\cos^6 x = 1 + \cos 2x = 2\cos^2 x$$

$$\Rightarrow \cos x = 0 \text{ or } \cos^2 x = \frac{2}{3}$$

$$\cos x = 0, \cos x = \pm \sqrt{\frac{2}{3}}$$

$\cos x = 0$ has 4 solutions.

$$\cos x = \pm \sqrt{\frac{2}{3}} \text{ has 8 solutions.}$$

Hence total 12 solutions.

9. Ans (1)

$$y = \frac{1}{2x-1} \Rightarrow x = \frac{1+y}{2y}$$

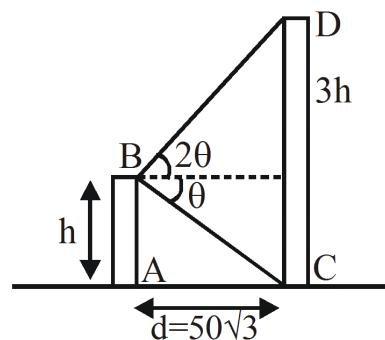
$\Rightarrow$ equation whose roots are $\frac{1}{2\alpha-1}, \frac{1}{2\beta-1}$ are

$$a\left(\frac{1+x}{2x}\right)^2 + b\left(\frac{1+x}{2x}\right) + c = 0$$

absolute term of above equation is 'a'

$$\Rightarrow C = a \Rightarrow \frac{C}{a} = 1$$

10. Ans (200)



$$\tan \theta = \frac{h}{d} \text{ \& \; } \tan 2\theta = \frac{3h}{d}$$

$$\Rightarrow \tan 2\theta = 3 \tan \theta$$

$$\frac{2}{1 - \tan^2 \theta} = 3$$

$$\Rightarrow \tan \theta = \frac{1}{\sqrt{3}} \text{ or } \theta = \frac{\pi}{6}$$

$$\Rightarrow h = 50\sqrt{3} \cdot \tan \frac{\pi}{6} = 50$$

$$CD = 4h = 200\text{m}$$